

Prismatic: The Adaptive Cognitive Engine

Refracting Complex Knowledge into 4 Personalized Learning Modalities on Amazon Quick

FOUNDING TEAM & AFFILIATION:

Saksham Kapoor (University of Maryland, College Park) · 10-Student Cross-Campus Builder Cohort

THE 300-STUDENT LECTURE HALL PARADOX:

One professor teaches 300 students in one rigid format. White light contains every color, but a flat chalkboard only projects one shade. When students freeze with intense brain fog, it's not a lack of intelligence—it's a cognitive mismatch.



SCAN FOR DEMO

GitHub repo, web app code,
prompt specs & cue cards.

< 30 SECONDS

Cognitive Triage

5-question diagnostic maps intake
profile

4 SPECTRUMS

Learning Modalities

Analogies, Schemas, Socratic &
Decks

2-SENTENCE CAP

Socratic Sparring

Active recall without text dumps

100% NO-CODE

Amazon Quick

Full multi-widget pipeline in 60
mins

The 4 Learning Spectrums: Zero Errors, Pure Retention

Orchestrated in a 60-minute build sprint on Amazon Quick & Amazon Bedrock foundation models

METAPHORICAL

Dorm & Dining Hall Realities

SPECTRUM 1: INTUITIVE ANALOGY

Translates abstract theory into daily campus navigation. E.g., Dijkstra's shortest path = navigating peak dining hall line rush at Blackburn Center.

VISUO-SPATIAL

Structural Concept Maps & ASCII

SPECTRUM 2: VISUAL SCHEMATIC

Renders execution trees, logical branches, and state machines into structural mental schemas for complex STEM and CS coursework.

KINESTHETIC

Active Retrieval Partner

SPECTRUM 3: SOCRATIC SPARRING

Chatbot strictly capped at 2 sentences; never lectures; fires progressive edge cases to force active recall without giving away answers.

SYNTHESIS

4-Slide Micro-Presentation

SPECTRUM 4: EXAM TRAP DECK

Distills the topic into 4 high-yield slides exposing the exact trick questions professors test on Midterm #2, paired with AI visual anchors.

Why Prismatic Wins: The \$80/Hour Tutoring Alternative

Comparing modern AI study workflows against passive text dumps and expensive human tutoring

VS. CHATGPT & RAW LLMS

Active Sparring vs. Text Dumps

ChatGPT dumps 800-word walls students glaze over. Prismatic caps replies at 2 sentences and forces active recall.

VS. \$80/HR PRIVATE TUTORING

Democratizing Academic Mastery

Tutors cost \$80/hr; office hours are 2 hrs/week. Prismatic gives every student 24/7 adaptive tutoring for pennies.

VS. STATIC SLIDES & SYLLABI

Dynamic Cognitive Fit

Ends task paralysis for ADHD, visual, and verbal learners with a 30-second diagnostic.

'With More Time We'd Add...' & The Shark Tank Verdict

Meeting 100% of official judging criteria: Creativity, Usefulness, Execution & Teamwork

CREATIVITY

Cognitive refraction metaphor
(white light → spectrums)

USEFULNESS

Directly solves dorm paralysis &
exam anxiety

EXECUTION

Working 6-widget pipeline on
Amazon Quick

TEAMWORK

10 teammates with dedicated active
roles

OFFICIAL HANDOUT MANDATE: 'WITH MORE TIME WE'D ADD...'

"With more time, we'd add live Canvas and syllabus LMS integration, automatically syncing weekly exam dates and generating proactive 15-minute micro-sprints before every morning class.

We don't just help students pass exams; we eliminate academic paralysis and unlock confidence."

WE ARE PRISMATIC — REFRACTING COMPLEX KNOWLEDGE INTO YOUR NATIVE SPECTRUM. VOTE PRISMATIC!